Pipeline and linepack formulation in AdOpT-NET0

`FluidynamicPipeline`: sets, parameters, variables and constraints

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This note writes out the optimization model that the class `FluidynamicPipeline` builds, read off the code at commit `9e530a28`. References in the right margin point to the code: `FP` = `fluidynamic_pipeline.py`, `FSP` = `fixed_size_pipeline.py` (both in `adopt_net0/components/networks/spe` `NW` = `networks/network.py`, `FL` = `networks/genericNetworks/fluid.py`. The reasoning behind each modelling choice is in `linepack_formulation.md` (decisions D1–D11).

Each pipeline type (one json in `network_data/`) is a separate network object with its own variables. Everything below is written for one such network; several types at the same node do *not* share variables (Section 6).

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1 Physics being approximated

Steady-state, isothermal, single-segment flow in a pipe $e = (i, j)$, hour t:

$$f_{e,t}\,|f_{e,t}| = R_e\,(p_{i,t}^2 - p_{j,t}^2) \qquad \text{pipeline (Weymouth-type) equation} \tag{1}$$

$$L_{e,t} = K_e\,\tfrac{1}{2}(p_{i,t} + p_{j,t}) \qquad \text{linepack} \tag{2}$$

$$L_{e,t} = L_{e,t-1} + \Delta t\,(f_{e,t}^{\text{in}} - f_{e,t}^{\text{out}}), \quad f_{e,t} = \tfrac{1}{2}(f_{e,t}^{\text{in}} + f_{e,t}^{\text{out}}) \qquad \text{pack balance} \tag{3}$$

Equations (2)–(3) are linear. Only (1) is not; it is replaced by a piecewise-linear (PWL) relation in Section 5.3. Flows are energy flows (MW) and the linepack is energy (MWh), consistent with the node energy balances.

2 Sets

Symbol	Code	Meaning
$t \in \mathcal{T}$	`set_t`	time steps
$a = (i, j) \in \mathcal{A}$	`set_arcs`, `arc_block[i,j]`	directed arcs; a bidirectional pipeline is two arcs (i, j) and (j, i)
$\bar{a} = (j, i)$		reverse arc of a
$e \in \mathcal{E}$	`set_arcs_unique`	pipelines (one per corridor)
$n \in \mathcal{N}^p$	`set_pressure_nodes`	pressure keys: the node if `pressure_coupled`= 1, one key per pipeline end if 0
$z \in \{0, \ldots, Z-1\}$	`set_breakpoints`	breakpoints of the PWL curve

3 Parameters

3.1 Input (json `Performance` and topology)

Symbol	json key	Meaning	Unit
D, ε	`diameter`, `roughness`	pipe diameter, wall roughness	m
$p^{\min}$, $p^{\max}$	`pressure_min`, `pressure_max`	pressure band	bar
p^{ref}	`pressure_ref`	level at which the PWL curve is built	bar
T, Z_c, M	`temperature`, `compressibility_factor`, `molar_mass`	gas state	K, –, kg/kmol
ed	`energy_density`	energy per mass	MWh/t
Z	`nr_breakpoints`	number of breakpoints (default 5)	–
L_a	topology	arc length	km
$\gamma_{1..4}$, …	economics	capex, opex, losses, compressor coefficients	

3.2 Derived per arc (`fit_network_performance`, `FSP:78`)

$$R_s = 8314/M \tag{4}$$

$$f_D = \left(2\log_{10}(3.7D/\varepsilon)\right)^{-2} \quad \text{Nikuradse, fully turbulent} \tag{5}$$

$$R_a = \frac{ed^2 \cdot 3.6^2 \cdot \frac{\pi^2}{16} D^5 \cdot 10^{10}}{f_D \cdot 1000 \cdot Z_c R_s T} \cdot \frac{1}{L_a} \quad [\text{MW}^2/\text{bar}^2] \tag{6}$$

$$K_a = ed \cdot \frac{\pi D^2}{4} \cdot 1000 \cdot \frac{100}{Z_c R_s T} \cdot L_a \quad [\text{MWh/bar}] \tag{7}$$

$$B_z = \frac{z}{Z-1}\left(p^{\max} - p^{\text{ref}}\right), \quad z = 0, \ldots, Z-1 \quad \text{drop breakpoints} \tag{8}$$

$$F_{a,z} = \sqrt{R_a\, B_z\, (2p^{\text{ref}} + B_z)} \quad \text{flow breakpoints} \tag{9}$$

$$\bar{F}_a = \max_z F_{a,z} \quad \text{arc capacity (json } \texttt{size_max} \text{ ignored)} \tag{10}$$

$F_{a,z}$ follows from (1) with the low pressure fixed at p^{ref} and the high one at $p^{\text{ref}} + B_z$ (decision D1). If `linepack_on`= 0, then $K_a := 0$ (`FP:202`).

3.3 Derived constants used in the constraints

$$\Delta P^{\text{act}} = \max_z B_z = p^{\max} - p^{\text{ref}}, \qquad \Delta P^{\text{unb}} = (p^{\max} - p^{\min}) - \Delta P^{\text{act}}, \tag{11}$$

Δt = `nr_timesteps_averaged` (1 without time staging), H = hours per typical day (0 without typical days), ρ = `rated_capacity` (1, so sizes and flows are in MW).

4 Variables

Symbol	Code	Domain	Meaning
Per arc a			
y_a	`var_installed`	$\{0,1\}$	arc built
s_a	`var_size`	≥ 0	size $= y_a \bar{F}_a$
$f_{a,t}$	`var_flow`	$[0, \bar{F}_a\rho]$	mean flow in the pipe
$f^{\text{in}}_{a,t}$, $f^{\text{out}}_{a,t}$	`var_flow_in/out`	$[0, \bar{F}_a\rho]$	flow entering / leaving
$\ell_{a,t}$	`var_losses`	$[0, \bar{F}_a\rho]$	transport losses
$\xi_{a,t}$	`var_direction`	$\{0,1\}$	direction a active
$\delta_{a,t}$	`var_delta_pressure`	$[0, \Delta P^{\text{act}}]$	pressure drop along a
$\lambda_{a,t,z}$	`var_lambda`	$[0,1]$	PWL weights (SOS2 over z)
Per pipeline e			
$L_{e,t}$	`var_linepack`	$[0, K_e p^{\max}]$	stored energy (MWh)
Per pressure key n			
$p_{n,t}$	`var_pressure`	$[p^{\min}, p^{\max}]$	pressure (bar)

Notation: for an arc $a = (i, j)$, $p_{i,t}$ is the pressure of node i *as seen by arc* a (`_get_pressure`, `FP:271`). With `pressure_coupled`= 1 all arcs of the network meeting at i see the same variable. Per pipeline and hour there are 2 binaries ($\xi_a, \xi_{\bar{a}}$) and 2 SOS2 sets.

5 Constraints

All constraints hold for every $t \in \mathcal{T}$ unless stated otherwise.

5.1 Investment (per arc)

$$s_a = y_a \, \bar{F}_a \qquad \texttt{FSP:189} \quad \text{(C1)}$$

$$y_a = 1 \quad \text{if the arc is existing} \qquad \texttt{FSP:187} \quad \text{(C2)}$$

$$\text{capex}_a = \gamma_1 y_a + \gamma_2 s_a + \gamma_3 L_a y_a + \gamma_4 s_a L_a \quad (= 0 \text{ if existing}) \qquad \texttt{FSP:210} \quad \text{(C3)}$$

$$y_a = y_{\bar{a}}, \qquad s_a = s_{\bar{a}} \qquad \texttt{FP:552, NW:858} \quad \text{(C4–C5)}$$

5.2 Flows on an arc

$$f_{a,t} = \tfrac{1}{2}(f^{\text{in}}_{a,t} + f^{\text{out}}_{a,t}) \qquad \texttt{FP:374} \quad \text{(C6)}$$

$$f^{\text{in}}_{a,t} \leq \xi_{a,t} \, \bar{F}_a \rho, \qquad f^{\text{out}}_{a,t} \leq \xi_{a,t} \, \bar{F}_a \rho \qquad \texttt{FP:382, FP:392} \quad \text{(C7–C8)}$$

$$f_{a,t} \leq s_a \rho \qquad \texttt{NW:785} \quad \text{(C9)}$$

$$f_{a,t} + f_{\bar{a},t} \leq s_a \rho \qquad \texttt{NW:871} \quad \text{(C10)}$$

$$\ell_{a,t} = f_{a,t} \cdot \text{loss} \cdot L_a \qquad \texttt{NW:776} \quad \text{(C11)}$$

The GDP disjunction of the parent class (`bidirectional_network_precise`) is switched off (`FP:185`); C7, C8 and C17 enforce the same condition.

5.3 Pipeline equation: piecewise linear with SOS2 (per arc)

$$\sum_z \lambda_{a,t,z} = \xi_{a,t} \qquad \texttt{FP:706} \quad \text{(C12)}$$

$$\delta_{a,t} = \sum_z B_z \, \lambda_{a,t,z} \qquad \texttt{FP:714} \quad \text{(C13)}$$

$$f_{a,t} = \sum_z F_{a,z} \, \lambda_{a,t,z} \qquad \texttt{FP:724} \quad \text{(C14)}$$

$$(\lambda_{a,t,0}, \ldots, \lambda_{a,t,Z-1}) \in \mathrm{SOS2} \qquad \texttt{FP:735} \quad \text{(C15)}$$

SOS2: at most two *adjacent* weights are non-zero, so (δ, f) lies on one segment of the curve. Because the weights sum to ξ (not to 1), an inactive direction has $\lambda = 0$, hence $\delta = 0$ and $f = 0$. Note that C14 interpolates the *mean* flow f.

5.4 Pressure drop and node pressures (per arc $a = (i, j)$)

$$\delta_{a,t} \le (p_{i,t} - p_{j,t}) + \Delta P^{\mathrm{act}}(1 - \xi_{a,t}) + \Delta P^{\mathrm{unb}}(1 - y_a) \qquad \texttt{FP:675} \quad \text{(C16a)}$$

$$\delta_{a,t} \ge (p_{i,t} - p_{j,t}) - \Delta P^{\mathrm{act}}(1 - \xi_{a,t}) - \Delta P^{\mathrm{unb}}(1 - y_a) \qquad \texttt{FP:690} \quad \text{(C16b)}$$

Built and active: $\delta_{a,t} = p_{i,t} - p_{j,t} \ge 0$, gas flows from high to low pressure. Inactive: $|p_i - p_j| \le \Delta P^{\mathrm{act}}$. Not built: the end pressures are free over the whole band. The big-M is split in two terms (decision D5) to keep it as small as possible on built pipelines.

5.5 Directions and pressures on an idle pipeline (per pipeline $e = (i, j)$)

$$\xi_{a,t} + \xi_{\bar{a},t} \le y_a \qquad \texttt{FP:572} \quad \text{(C17)}$$

$$p_{i,t} - p_{j,t} \le \Delta P^{\mathrm{act}}(\xi_{a,t} + \xi_{\bar{a},t} + 1 - y_a) + \Delta P^{\mathrm{unb}}(1 - y_a) \qquad \texttt{FP:598} \quad \text{(C18a)}$$

$$p_{i,t} - p_{j,t} \ge -\Delta P^{\mathrm{act}}(\xi_{a,t} + \xi_{\bar{a},t} + 1 - y_a) - \Delta P^{\mathrm{unb}}(1 - y_a) \qquad \texttt{FP:611} \quad \text{(C18b)}$$

A built pipeline may be idle ($\xi_a = \xi_{\bar{a}} = 0$); C18 then forces $p_{i,t} = p_{j,t}$.

5.6 Linepack (per pipeline $e = (i, j)$)

With $\bar{p}_{e,t} = \frac{1}{2}(p_{i,t} + p_{j,t})$:

$$L_{e,t} \le K_e \, p^{\max} \, y_e \qquad \texttt{FP:453} \quad \text{(C19)}$$

$$L_{e,t} \ge K_e \, p^{\min} \, y_e \qquad \texttt{FP:466} \quad \text{(C20)}$$

$$L_{e,t} \le K_e \, \bar{p}_{e,t} \qquad \texttt{FP:483} \quad \text{(C21)}$$

$$L_{e,t} \ge K_e \, \bar{p}_{e,t} - K_e \, p^{\max}(1 - y_e) \qquad \texttt{FP:499} \quad \text{(C22)}$$

$$L_{e,t} = L_{e,\mathrm{prev}(t)} + \Delta t \sum_{d \in \{a, \bar{a}\}} (f^{\mathrm{in}}_{d,t} - f^{\mathrm{out}}_{d,t}) \qquad \texttt{FP:513} \quad \text{(C23)}$$

Built: C21–C22 give $L = K\bar{p}$, i.e. (2). Not built: C19 gives $L = 0$. The predecessor is

$$\mathrm{prev}(t) = \begin{cases} t - 1, & \text{otherwise,} \\ |\mathcal{T}|, & t = 1 \text{ (no typical days: cyclic over the horizon),} \\ t + H - 1, & (t - 1) \bmod H = 0 \text{ (typical days, method 1: cyclic per day).} \end{cases}$$

No initial linepack is imposed.

5.7 Coupling to the rest of the model (per node n)

$$\text{inflow}_{n,t} = \sum_{i \to n} \left(f^{\text{out}}_{(i,n),t} - \ell_{(i,n),t} \right) \qquad \texttt{FP:753} \quad \text{(C24)}$$

$$\text{outflow}_{n,t} = \sum_{n \to j} f^{\text{in}}_{(n,j),t} \qquad \texttt{FP:773} \quad \text{(C25)}$$

$$\text{cons}^{\text{send}}_{a,t} = f_{a,t} \left(k^{\text{flow}} + k^{\text{flowDist}} L_a \right) \quad \text{(same at the receiving end)} \qquad \texttt{FL:212} \quad \text{(C26)}$$

$$\text{opex}^{\text{var}} \propto \textstyle\sum_{a,t} f_{a,t}, \qquad \text{opex}^{\text{fix}} \propto \text{capex} \qquad \texttt{NW:812} \quad \text{(C27)}$$

The node balance sees the end flows, so gas being packed is not delivered. Losses, compressor energy and variable opex see the mean flow. The compressor coefficients k are computed from the fixed json pressures and do not depend on $p_{n,t}$ (decision D8).

6 How pressures connect in a network

Three ingredients together act like Kirchhoff's laws:

1. **Current law** (node balance, C24–C25 in the ADOPT energy balance): what enters a node equals what leaves it. The reference transport model already has this law.
2. **Potential**: on every active arc $\delta_a = p_i - p_j$ (C16), with p_i one variable per node, shared by every arc of the network at i.
3. **Pipe law**: $f_a = \text{PWL}(\delta_a)$ (C12–C15).

Around any loop $\sum(p_i - p_j) = 0$ identically, so $\sum_{\text{loop}} \pm\delta_a = 0$ and, through the pipe law, $\sum_{\text{loop}} \pm\text{PWL}^{-1}(f_a) = 0$: the *voltage law*. It is not written explicitly; it follows from point 2. As a consequence, the split of flow between parallel paths is fixed by the physics instead of being free.

Which pressure does an arc see at node n?

Arcs meeting at n	Pressures at n
same pipeline type, `pressure_coupled`= 1 (default)	one, shared
different pipeline types	one per type, not linked
same type, `pressure_coupled`= 0	one per pipeline end

Caveat (D9): `medium` and `large` share the band 15–30 bar but get separate pressures at the same node. A fix (R2) is $p^{\text{medium}}_{n,t} = p^{\text{large}}_{n,t}$ for every shared node. Between the backbone (30–70 bar) and the distribution grid the pressures are independent by design, which acts as a free regulator or compressor.

7 Tightened variants (`examples/ladder/linepack_formulations.py`)

Applied after the model is built; same integer feasible set, tighter LP relaxation. $D = p^{\max} - p^{\min}$.

Variant	Removes	Adds
`hull`	C21	$L \le K\bar{p} - Kp^{\min}(1-y)$
`signed`	C16, C18	$\lvert(p_i - p_j) - (\delta_a - \delta_{\bar{a}})\rvert \le D(1-y)$
`combined`	both of the above	both of the above
`combined_cuts`	as `combined`	$Kp^{\min}y + \frac{K}{2}(\delta_a + \delta_{\bar{a}}) \le L \le Kp^{\max}y - \frac{K}{2}(\delta_a + \delta_{\bar{a}})$
`magnitude_cuts`	`combined_cuts`, C12–C15 of both arcs	one SOS2 per pipeline: $\sum_z \lambda_{e,t,z} = y$, $f_a + f_{\bar{a}} = \sum_z F_z\lambda$, $\delta_a + \delta_{\bar{a}} = \sum_z B_z\lambda$; $\xi_a + \xi_{\bar{a}} = y$ with $\xi_{\bar{a}}$ continuous; $\delta_d \le \Delta P^{\text{act}}\xi_d$
`magnitude_ratio_cuts`	as `magnitude_cuts`	$r^{\min}\delta_d \le f_d \le r^{\max}\delta_d$ per direction d, $r = F_z/B_z$ over $B_z > 0$

8 Possible extensions (not implemented)

R1, inequality relaxation. The curve $F(\Delta p)$ is concave, so the PWL equals the minimum of its secant lines, $s_k = \frac{F_{k+1}-F_k}{B_{k+1}-B_k}$, $b_k = F_k - s_k B_k$. Replacing C12–C15 by

$$f_{a,t} \le s_k\,\delta_{a,t} + b_k\,\xi_{a,t} \qquad k = 0, \dots, Z-2$$

describes exactly the region *under* the curve without λ or SOS2. It gives up $f \ge F(\Delta p)$, i.e. it allows throttling. The direction binaries ξ must stay: without them both arcs of a pipeline would force $p_i \ge p_j$ and $p_j \ge p_i$, hence no flow.

Gas velocity limit. With $\rho = p/(Z_c R_s T)$ and $\dot{m} = f/ed$, the velocity is $v = \dot{m}/(\rho A)$, $A = \pi D^2/4$. A cap $v \le v^{\max}$ is linear in flow and pressure:

$$f^{\text{in}}_{a,t} \le c\,v^{\max}\,p_{i,t}, \qquad f^{\text{out}}_{a,t} \le c\,v^{\max}\,p_{j,t}, \qquad c = \frac{ed \cdot A}{Z_c R_s T} \text{ (with unit conversion).}$$

At capacity the current model allows 17–35 m/s at the inlet and up to 52–81 m/s if the pipe runs at $p^{\min}$, so a realistic cap would bind.

9 What is not modelled

- dependence of the flow on the absolute pressure level (1-D curve at p^{ref}, D1);
- spatial discretisation, gas inertia, temperature dynamics;
- compressor work as a function of the pipeline pressure (D8);
- a shared pressure across pipeline types of the same band (D9);
- at most one pipeline type per corridor (R3);
- an initial linepack (the balance is cyclic).